

Introduction to Encryption and Security Motivation

Introduction to Encryption

Encryption is one of the most important techniques used in information security to protect digital information from unauthorized access.

In today's digital world, large amounts of sensitive information such as passwords, financial details, personal messages, and business data are transferred through networks.

During communication, attackers may try to capture or modify this information. Encryption provides protection by converting readable information into an unreadable format so that only authorized users can access the original data.

Why Do We Need Encryption?

Encryption is required because information transmitted through the internet can be exposed to various security threats.

1



Protection of
confidential
information

2



Prevention of
unauthorized
access

3



Secure
communication
over networks

4



Protection of
online
transactions

5

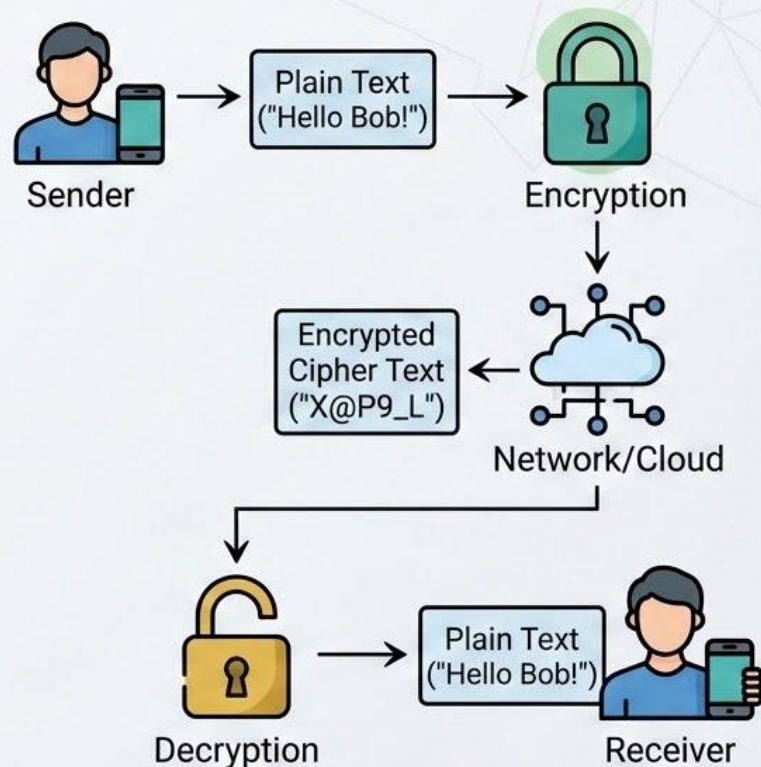


Maintaining
user privacy
and trust

Real Life Applications of Encryption

WhatsApp Message Security

- Ensures private conversations remain accessible **ONLY** to the sender and receiver.
- Converts messages into encrypted form during transmission and decrypts them back into readable **form** at the receiver side.



Real-Life Applications of Encryption



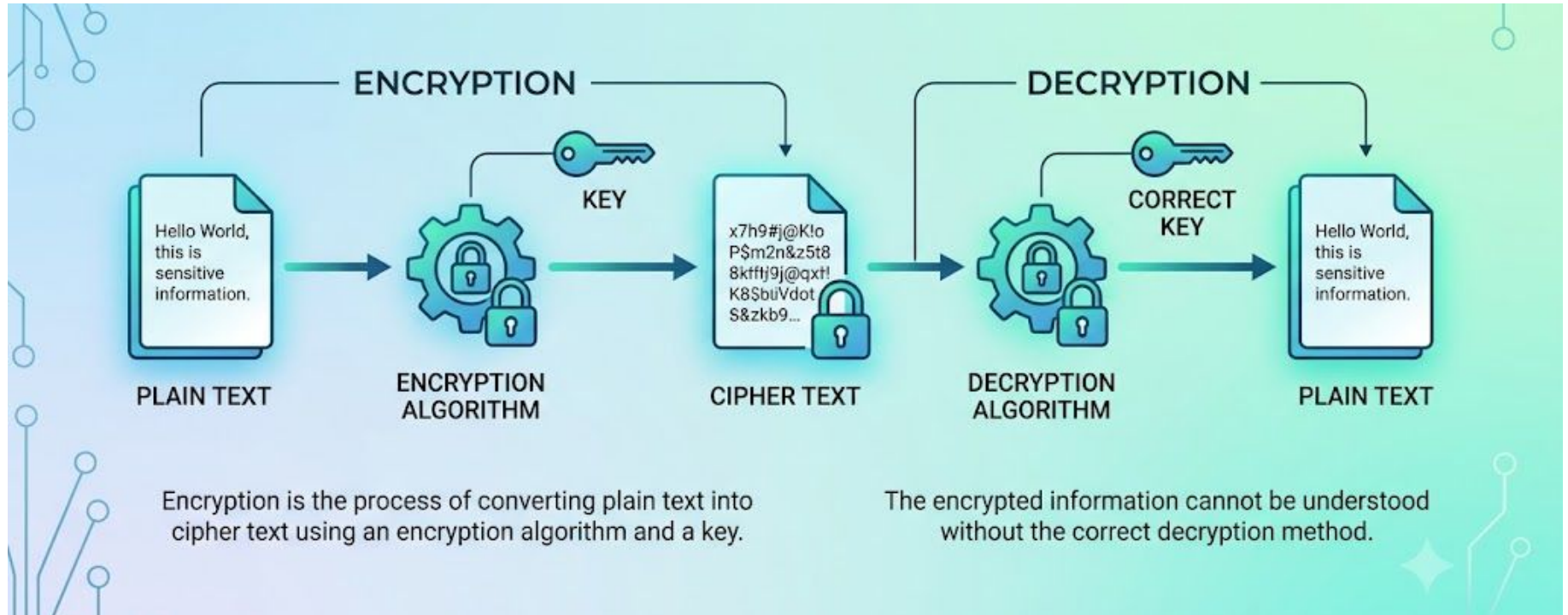
Online Banking Security

Encryption protects sensitive information such as usernames, passwords, and transaction details from attackers.



ATM PIN Security ATM machines and banking systems use encryption to prevent PIN theft. Encryption protects communication between ATM machines and banking systems to prevent PIN theft.

Basic Encryption Concept



Plain Text

- Plain text is the **original readable information** before applying encryption.
- It is the input given to the encryption algorithm.
- Example: **HELLO**
- The sender creates plain text before sending information to the receiver.

Cipher Text

- Cipher text is the **encrypted and unreadable form of information generated after encryption.**
- It protects the original message from unauthorized users.

Example:

- **Plain Text:** HELLO
- **Cipher Text:** KHOOR

Encryption Algorithm

- An encryption algorithm is a **mathematical process used to convert plain text into cipher text**.
- The algorithm applies specific rules and operations using a key to transform the original message.

Examples:

- AES - Advanced Encryption Standard
- DES - Data Encryption Standard
- RSA - Rivest–Shamir–Adleman
- Caesar Cipher

Decryption Algorithm

- Decryption is the reverse process of encryption.
- The decryption algorithm converts cipher text back into the original plain text.
- Only users having the correct key can successfully decrypt the information.

Simple Encryption Example

Caesar Cipher is a basic encryption technique where each character is shifted by a fixed number.

Example:

Message: HELLO

Encryption Rule: Shift +3

Result: KHOOR

The receiver applies reverse shifting to recover the original message.

Security Goals of Encryption

Encryption provides three major security services:

- Confidentiality
- Integrity
- Authentication

These are known as the basic principles of information security.

Confidentiality

- Confidentiality ensures that information is accessible only to authorized users.
- It prevents unauthorized persons from viewing sensitive information.

Example: Online banking passwords should be visible only to the user and bank system.

Methods:

- Encryption
- Access control
- Authentication

Integrity

- Integrity ensures that information remains accurate and is not modified during transmission.
- The receiver should receive the same information that was sent by the sender.

Example: An online transaction amount should not be changed by attackers.

Methods:

- Hashing
- Digital signatures
- Encryption

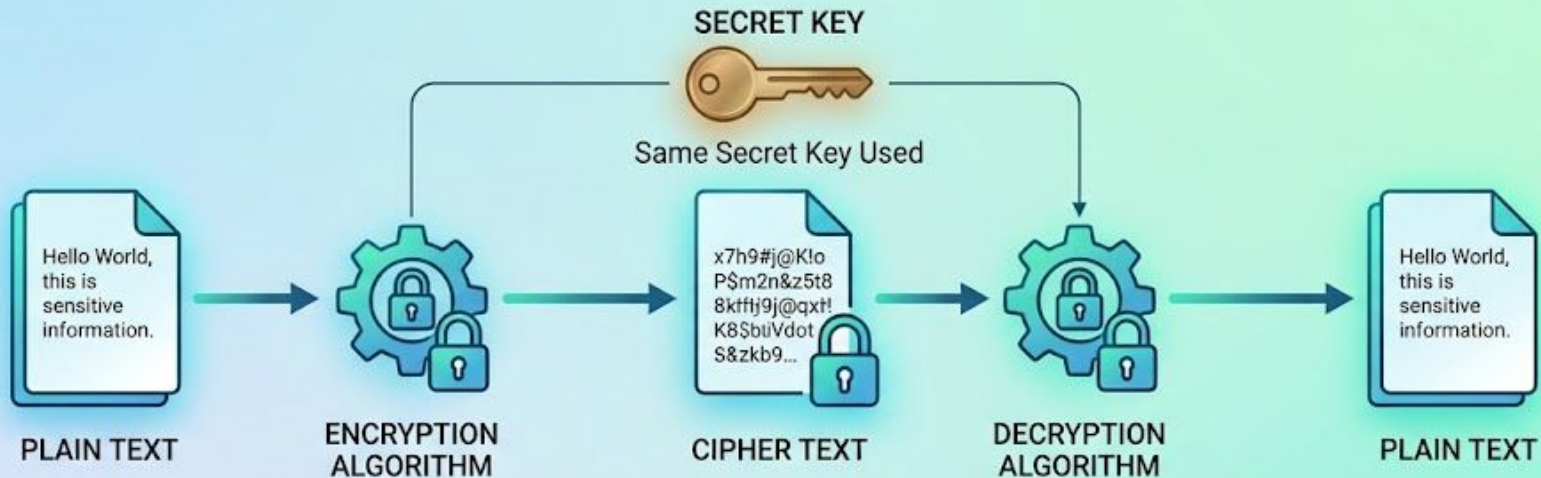
Authentication

- Authentication verifies the identity of users or systems involved in communication.
- It ensures that the information is exchanged between trusted parties.

Examples:

- Password verification
- OTP authentication
- Biometric authentication
- Digital certificates

CONVENTIONAL ENCRYPTION MODEL



Conventional encryption is also known as symmetric encryption.

It uses the same secret key for both encryption and decryption. The sender and receiver must share the same key securely.

Components of Conventional Encryption Model

Sender

The sender creates the original message and applies encryption before transmission.

Receiver

The receiver receives encrypted information and performs decryption to obtain the original message.

Encryption Algorithm

A process that converts plain text into cipher text.

Decryption Algorithm

A process that converts cipher text into original plain text.

Secret Key

A confidential key used for encryption and decryption.

Attacker

An unauthorized person who attempts to access, modify, or steal information.

Working of Conventional Encryption

Step 1:

Sender creates plain text message.

Step 2:

Encryption algorithm uses secret key to convert plain text into cipher text.

Step 3:

Cipher text is transmitted through the network.

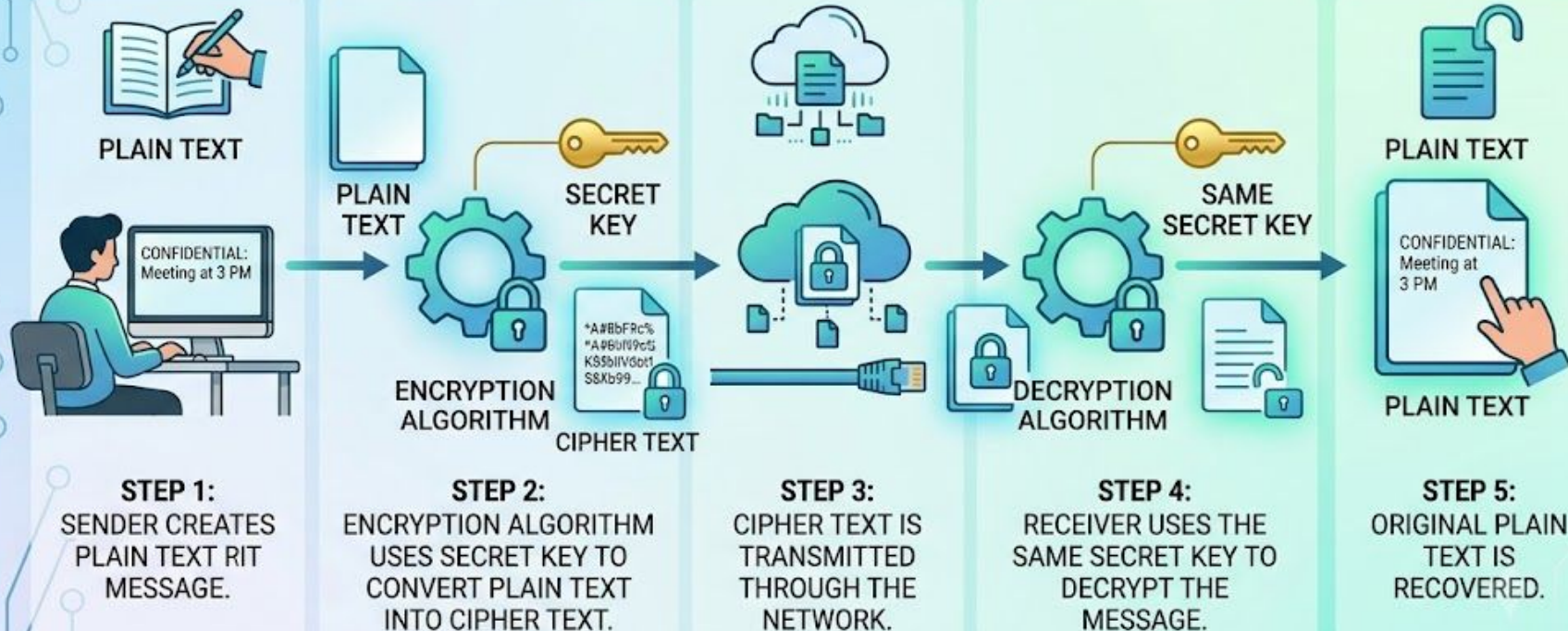
Step 4:

Receiver uses the same secret key to decrypt the message.

Step 5:

Original plain text is recovered.

WORKING OF CONVENTIONAL ENCRYPTION (STEP-BY-STEP)



Advantages of Conventional Encryption

- Provides fast encryption and decryption.
- Suitable for protecting large amounts of data.
- Requires less computational power.
- Widely used for data protection.

Limitations of Conventional Encryption

- Secure key sharing is required between sender and receiver.
- If the secret key is stolen, encrypted information can be accessed.
- Managing multiple keys becomes difficult in large networks.

Summary

Encryption protects information by converting plain text into cipher text.

It provides:

- Confidentiality
- Integrity
- Authentication

Conventional encryption uses the same secret key for encryption and decryption.

It is an important foundation of modern information security systems.

Steganography

Introduction to Steganography

- Steganography is a technique of hiding secret information inside another normal-looking file or object.
- The main purpose of steganography is to hide the existence of communication.
- Unlike encryption, where the message becomes unreadable, **steganography hides the message itself** so that others do not know that any secret communication exists.

Examples of cover objects:

- Images
- Audio files
- Video files
- Text documents

Encryption vs Steganography

Encryption

Encryption hides the meaning of information.

The message is visible but appears unreadable.

Example:

Original Message: "Meet at 5 PM"

Encrypted Message: "X7#P9@L"

An attacker knows that a secret message exists but cannot understand it.

Steganography

Steganography hides the existence of information.

The file appears normal, and others may not know that a message is hidden inside it.

Example:

Normal Image: Cat.jpg

Hidden Message: "Meet at 5 PM"

The image looks unchanged, but the message is secretly stored inside it.

BASIC CONCEPT OF STEGANOGRAPHY

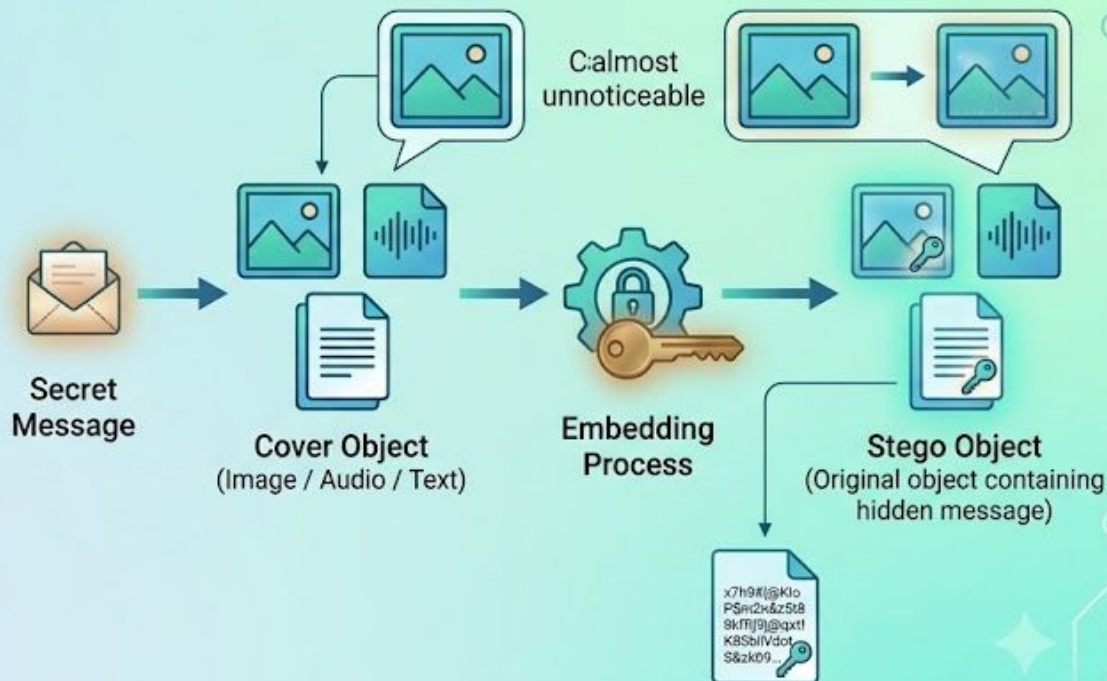
Steganography works by embedding secret information inside a cover object.



- The original file used for hiding information is called the **cover object**.



- After hiding the message, the resulting file is called the **stego object**.



Important Terms in Steganography

Secret Message

The information that needs to be hidden.

Examples:

- Text message
- Password
- Confidential document

Cover Object

The normal file used to hide the secret information.

Examples:

- Image
- Audio
- Video
- Text file

Stego Object

- The final object after embedding the secret message.
- It looks similar to the original file but contains hidden information.

HOW STEGANOGRAPHY WORKS (STEP-BY-STEP)

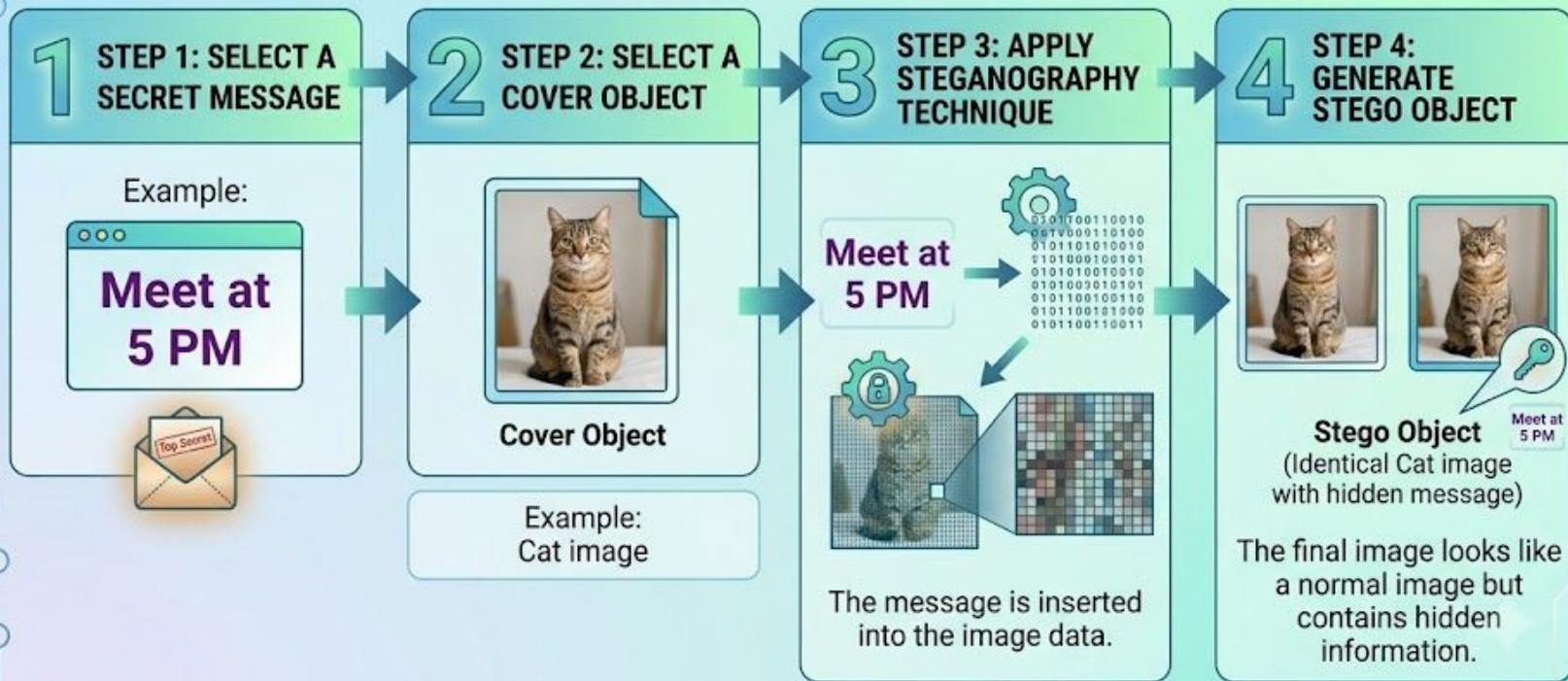


IMAGE STEGANOGRAPHY

OVERVIEW

Image steganography is the most commonly used type of steganography.



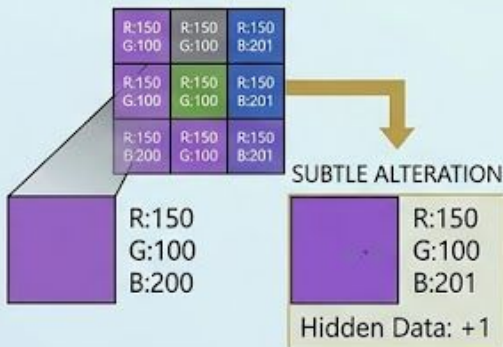
It hides information inside digital images by modifying image pixels.



It hides information inside digital images by modifying image pixels.

PIXELS AND VISUAL PERCEPTION

- Digital images are made up of small units called **pixels**.
- Each pixel contains color information.
- Small changes in pixel values can store hidden information without noticeable visual changes.



STEGANOGRAPHY EXAMPLE

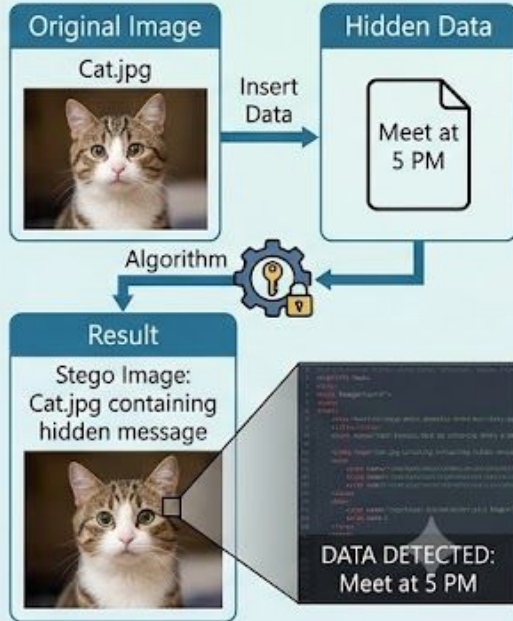
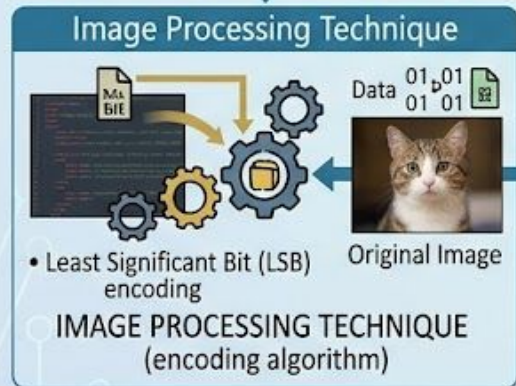


IMAGE STEGANOGRAPHY PROCESSING WORKFLOW & EXTRACTION

HIDING DIAGRAM



EXTRACTION DIAGRAM



Stego Image Input



The receiver extracts the hidden message from the stego image using the correct extraction method.

Example of Hidden Message in Image

Normal Image: A picture of a cat

Question: Can you see any hidden information?

Answer: No visible change appears in the image.

Internally:

Image pixels contain hidden text information.

The message can only be extracted using a steganography tool or method.

Types of Steganography

Image Steganography

Secret data is hidden inside images.

Example:

JPEG, PNG images

Audio Steganography

Secret information is hidden inside audio files.

Example:

MP3 files

Video Steganography

Information is hidden inside video frames.

Example:

MP4 videos

Text Steganography

Information is hidden inside text documents.

Example:

Using spaces or character patterns

Techniques Used in Image Steganography

Least Significant Bit (LSB)

- LSB is one of the most common image steganography techniques.
- The least important bits of image pixels are modified to store secret information.
- Since the change is very small, the human eye cannot easily detect the difference.

Example:

Original Pixel:

10110110

Modified Pixel:

10110111

Only the last bit changes, but it can store information.

Applications of Steganography

Military Communication

Used for secure communication where the existence of messages must remain hidden.

Example: Secret information sent through images.

Digital Watermarking

Used to protect digital content ownership.

Example: Adding hidden copyright information to images, videos, and documents.

Secret Communication

Used for private communication where users want to avoid detection.

Digital Watermarking

Digital watermarking is a technique where information about ownership or authenticity is embedded into digital content.

Applications:

- Copyright protection
- Document verification
- Media authentication

Example: A photographer embeds ownership information inside an image.

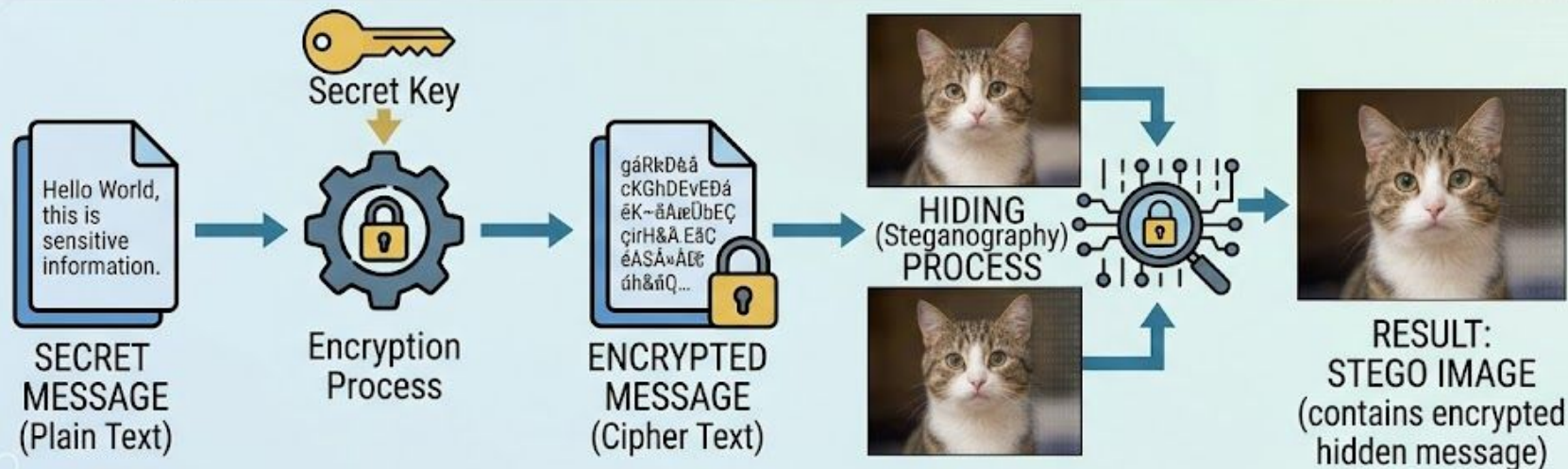
Advantages of Steganography

- Hides the existence of communication.
- Provides an additional security layer.
- Useful for protecting confidential information.
- Can be combined with encryption for stronger security.

Limitations of Steganography

- Hidden information capacity is limited.
- Large modifications may make the file suspicious.
- Advanced analysis techniques can detect hidden data.
- Loss of quality may occur during file compression.

ENCRYPTION AND STEGANOGRAPHY TOGETHER



The secret message is first encrypted.

The encrypted message is then hidden inside the cover image.

Even if attackers find the hidden message, they cannot understand it without the encryption key.

Classroom Activity

Activity: Show students two images:

Image 1: Normal image

Image 2: Image containing hidden text

Ask:

"Can you see the hidden message?"

Discussion:

The message is not visible because steganography hides information inside the file structure.

Summary

Steganography is a technique used to hide the existence of secret information.

Encryption hides the meaning of information.

Steganography hides the presence of information.

Main components:

- Secret Message
- Cover Object
- Stego Object

Applications:

- Military communication
- Digital watermarking
- Secret communication
- Steganography provides an additional security layer for protecting digital information.

Thank you.!